Supply Chain Optimization

Project Description – Supply Chain Optimization :

This project focuses on designing and implementing a relational database system for supply chain optimization using MySQL. The system models key entities involved in the supply chain, including suppliers, manufacturers, distributors, products, inventory, orders, and logistics. Each table is structured with appropriate primary keys, foreign keys, and constraints to maintain data integrity and establish relationships between different components of the supply chain. The database enables efficient storage and retrieval of data related to product flow, stock levels, and order processing.

Furthermore, SQL queries are implemented to perform essential operations such as identifying low inventory levels for timely replenishment, analyzing total orders handled by distributors, evaluating transportation costs, and tracking order fulfillment status. These queries help in minimizing stock shortages, reducing logistics expenses, and improving overall operational efficiency. The project demonstrates how a well-structured database system can support decision-making and optimize supply chain management in real-world scenarios.

CREATE DATABASE :

```
CREATE DATABASE supply_chain_db;

USE supply_chain_db;
```

Output

Action Output

#	Time	Action	Message
1	09:33:40	CREATE DATABASE supply_chain_db	1 row(s) affected
2	09:33:40	USE supply_chain_db	0 row(s) affected

CREATE TABLE :

➔ **Suppliers Table :**

```
CREATE TABLE suppliers (
  supplier_id INT PRIMARY KEY AUTO_INCREMENT,
  name VARCHAR(100) NOT NULL,
  contact VARCHAR(15),
  city VARCHAR(50)
);
```

3	09:37:22	CREATE TABLE suppliers (	supplier_id INT PRIMARY KEY AUTO_INCREMENT,	name VARCHAR(100) N...	0 row(s) affected

➔ **Manufacturers Table :**

```
CREATE TABLE manufacturers (
  manufacturer_id INT PRIMARY KEY AUTO_INCREMENT,
  name VARCHAR(100) NOT NULL,
  location VARCHAR(50)
);
```

4	09:41:53	CREATE TABLE manufacturers (manufacturer_id INT PRIMARY KEY AUTO_INCREMENT, name VARCH...	0 row(s) affected

➔ **Distributors Table :**

```
CREATE TABLE distributors (
  distributor_id INT PRIMARY KEY AUTO_INCREMENT,
  name VARCHAR(100) NOT NULL,
  region VARCHAR(50)
);
```

5	09:43:08	CREATE TABLE distributors (distributor_id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(100)...	0 row(s) affected

➔ **Products Table :**

```
CREATE TABLE products (
  product_id INT PRIMARY KEY AUTO_INCREMENT,
  name VARCHAR(100) NOT NULL,
  manufacturer_id INT,
  FOREIGN KEY (manufacturer_id) REFERENCES manufacturers(manufacturer_id)
);
```

6	09:45:20	CREATE TABLE products (product_id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(100) N...	0 row(s) affected

➔ **Inventory Table :**

```
CREATE TABLE inventory (
  inventory_id INT PRIMARY KEY AUTO_INCREMENT,
  product_id INT,
  quantity INT CHECK (quantity >= 0),
  location VARCHAR(50),
  FOREIGN KEY (product_id) REFERENCES products(product_id)
);
```

7	09:46:41	CREATE TABLE inventory (inventory_id INT PRIMARY KEY AUTO_INCREMENT, product_id INT, quan...	0 row(s) affected

➔ **Orders Table :**

```
CREATE TABLE orders (
  order_id INT PRIMARY KEY AUTO_INCREMENT,
  product_id INT,
  distributor_id INT,
  quantity INT,
  order_date DATE,
  FOREIGN KEY (product_id) REFERENCES products(product_id),
  FOREIGN KEY (distributor_id) REFERENCES distributors(distributor_id)
);
```

✔	8	09:47:50	CREATE TABLE orders (	order_id INT PRIMARY KEY AUTO_INCREMENT,	product_id INT,	distributor_i...	0 row(s) affected

➔ **Logistics Table :**

```
CREATE TABLE logistics (
  logistics_id INT PRIMARY KEY AUTO_INCREMENT,
  order_id INT,
  transport_mode VARCHAR(50),
  cost DECIMAL(10,2),
  delivery_status VARCHAR(50),
  FOREIGN KEY (order_id) REFERENCES orders(order_id)
);
```

✔	9	09:48:51	CREATE TABLE logistics (	logistics_id INT PRIMARY KEY AUTO_INCREMENT,	order_id INT,	transport...	0 row(s) affected

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Insertion of Sample Data :

```
INSERT INTO suppliers (name, contact, city)
VALUES ('ABC Supplies', '9876543210', 'Delhi');

INSERT INTO manufacturers (name, location)
VALUES ('XYZ Manufacturing', 'Gurgaon');

INSERT INTO distributors (name, region)
VALUES ('Fast Distributors', 'North India');

INSERT INTO products (name, manufacturer_id)
VALUES ('Product A', 1);

INSERT INTO inventory (product_id, quantity, location)
VALUES (1, 100, 'Warehouse 1');

INSERT INTO orders (product_id, distributor_id, quantity, order_date)
```

VALUES (1, 1, 20, '2026-04-01');

INSERT INTO logistics (order_id, transport_mode, cost, delivery_status)
VALUES (1, 'Truck', 5000, 'In Transit');

10	09:52:12	INSERT INTO suppliers (name, contact, city) VALUES ('ABC Supplies', '9876543210', 'Delhi')	1 row(s) affected
11	09:52:12	INSERT INTO manufacturers (name, location) VALUES ('XYZ Manufacturing', 'Gurgaon')	1 row(s) affected
12	09:52:12	INSERT INTO distributors (name, region) VALUES ('Fast Distributors', 'North India')	1 row(s) affected
13	09:52:12	INSERT INTO products (name, manufacturer_id) VALUES ('Product A', 1)	1 row(s) affected
14	09:52:12	INSERT INTO inventory (product_id, quantity, location) VALUES (1, 100, 'Warehouse 1')	1 row(s) affected
15	09:52:12	INSERT INTO orders (product_id, distributor_id, quantity, order_date) VALUES (1, 1, 20, '2026-04-01')	1 row(s) affected
16	09:52:12	INSERT INTO logistics (order_id, transport_mode, cost, delivery_status) VALUES (1, 'Truck', 5000, 'In Transit')	1 row(s) affected

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Check Low Stock :

SELECT p.name, i.quantity

FROM inventory i

JOIN products p ON i.product_id = p.product_id

WHERE i.quantity < 50;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	quantity
Product A	100

Result 2 × Read Only

18	09:57:04	SELECT p.name, i.quantity FROM inventory i JOIN products p ON i.product_id = p.product_id WHERE i.quanti...	1 row(s) returned

Total Orders by Distributors :

SELECT d.name, SUM(o.quantity) AS total_orders

FROM orders o

JOIN distributors d ON o.distributor_id = d.distributor_id

GROUP BY d.name;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	total_orders
Fast Distributors	20

Result 3 × Read Only

19	09:59:10	SELECT d.name, SUM(o.quantity) AS total_orders FROM orders o JOIN distributors d ON o.distributor_id = d.di...	1 row(s) returned

Transportation Cost Analysis :

```
SELECT transport_mode, SUM(cost) AS total_cost
FROM logistics
GROUP BY transport_mode;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

transport_mode	total_cost
Truck	5000.00

Result 4 × Read Only

20 10:00:43 SELECT transport_mode, SUM(cost) AS total_cost FROM logistics GROUP BY transport_mode 1 row(s) returned

Order Fulfillment Status :

```
SELECT o.order_id, p.name, l.delivery_status
FROM orders o
JOIN products p ON o.product_id = p.product_id
JOIN logistics l ON o.order_id = l.order_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	name	delivery_status
1	Product A	In Transit

Result 5 × Read Only

21 10:02:05 SELECT o.order_id, p.name, l.delivery_status FROM orders o JOIN products p ON o.product_id = p.product_id... 1 row(s) returned